NurNabi Academy

IGCSE Mathematics · exam-style revision

IGCSE Revision Paper

NAME

CLASS / CENTRE

DATE

MATHEMATICS **NNA 0580/04**

Paper 4 Calculator (Extended) **October/November 2026**

Full Revision Paper 1 **2 hours**

You must answer on the question paper.

You will need: Geometrical instruments

INSTRUCTIONS

- Answer **all** questions.
- Use a black or dark blue pen. You may use an HB pencil for any diagrams or graphs.
- Write your name in the box at the top of the page.
- Write your answer to each question in the space provided.
- Do **not** use an erasable pen or correction fluid.
- You should use a scientific calculator where appropriate.
- You may use tracing paper.
- You must show all necessary working clearly.
- Give non-exact numerical answers correct to 3 significant figures, or 1 decimal place for angles in degrees, unless a different level of accuracy is specified in the question.
- For π, use either your calculator value or 3.142.

INFORMATION

- The total mark for this paper is **100**.
- This document has **19** pages.
- The number of marks for each question or part question is shown in brackets [ ].

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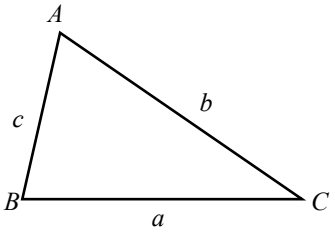
List of formulas

Area, A, of triangle, base b, height h.	$A = \frac{1}{2}bh$
Area, A, of circle of radius r.	$A = \pi r^2$
Circumference, C, of circle of radius r.	$C = 2\pi r$
Curved surface area, A, of cylinder of radius r, height h.	$A = 2\pi rh$
Curved surface area, A, of cone of radius r, sloping edge l.	$A = \pi rl$
Surface area, A, of sphere of radius r.	$A = 4\pi r^2$
Volume, V, of prism, cross-sectional area A, length l.	$V = Al$
Volume, V, of pyramid, base area A, height h.	$V = \frac{1}{3}Ah$
Volume, V, of cylinder of radius r, height h.	$V = \pi r^2 h$
Volume, V, of cone of radius r, height h.	$V = \frac{1}{3}\pi r^2 h$
Volume, V, of sphere of radius r.	$V = \frac{4}{3}\pi r^3$

For the equation $ax^2 + bx + c = 0$, where $a \neq 0$,

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

For the triangle shown,



$$\frac{a}{\sin A} = \frac{b}{\sin B} = \frac{c}{\sin C}$$

$$a^2 = b^2 + c^2 - 2bc\cos A$$

$$\text{Area} = \frac{1}{2}ab\sin C$$

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1 Solve.

$$4(3x - 5) = 2(x + 9) + 2$$

$x =$ ..... [3]

2 Work out.

$$\frac{16.71 + 46.13}{\sqrt{8.6^2 - 3.5^2}}$$

Give your answer correct to 2 significant figures.

..... [2]

3 In a sale, the price of a printer is reduced by 18% to \$287.

Work out the original price of the printer.

\$ ..... [2]

4 Amira, Bilal and Chen share \$4560 in the ratio 5 : 7 : 12.

Calculate the amount Chen receives.

\$ ..... [2]

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5 These are the first 5 terms of a sequence.

2 6 12 20 30

(a) Find the next term of this sequence.

..... [1]

(b) Find the nth term of this sequence.

..... [2]

6 Factorise fully.

$6x^2 - 7x - 3$

..... [2]

7 Amira invests \$950 at a rate of 3.4% per year compound interest.

Calculate the value of her investment at the end of 6 years.
Give your answer correct to the nearest dollar.

\$ ..... [3]

8 Bilal invests \$1330 at a rate of r% per year **simple** interest.
At the end of 8 years the total value of his investment is \$1968.40.

Calculate the value of r.

$r =$ ..... [3]

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9 Solve the simultaneous equations.
You must show all your working.

$3x + 2y = 4$
$5x - 4y = 25$

$x =$ ..... , $y =$ ..... [3]

10 Simplify fully.

$$\frac{3x^2 - 12}{x^2 - x - 6}$$

..... [3]

11 Make x the subject of this formula.

$$y = \frac{3x + 2}{x - 5}$$

$x =$ ..... [4]

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12 The interior angle of a regular polygon is 162°.

Find the number of sides of this polygon.

..... [2]

13 A curve has equation $y = x^3 - 6x^2 + 9x + 2$.

Find the coordinates of the two turning points of the curve and determine whether each one is a maximum or a minimum.
You must show all your working.

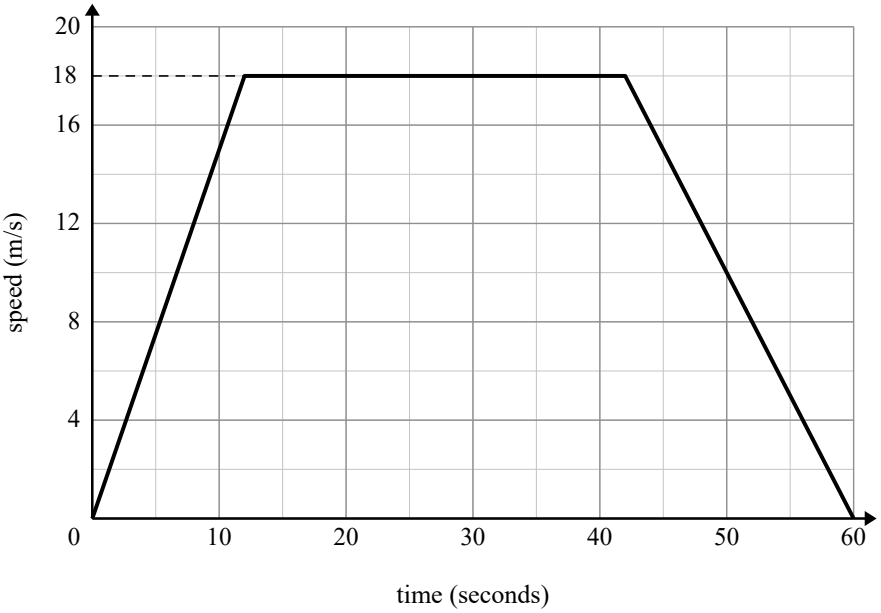
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(....., ...) ..... [5]

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14 The speed–time graph shows the journey of a car.



(a) Calculate the acceleration during the first 12 seconds.

..... m/s^2 [1]

(b) Calculate the total distance travelled.

..... m [3]

(c) Calculate the average speed for the whole journey.

..... m/s [1]

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15 A machine is worth \$24 000.
Its value decreases exponentially by 12% each year.

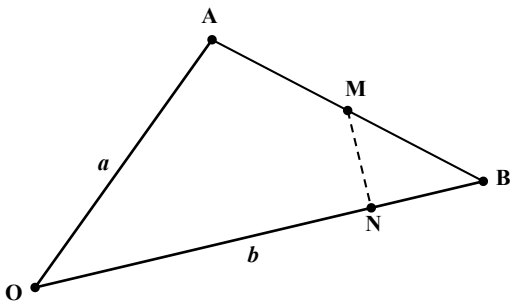
(a) Calculate the value of the machine after 5 years.

\$ ..... [2]

(b) Find the number of complete years before the value of the machine first falls below \$8000.

..... years [2]

16 In the diagram, **OA** = $\boldsymbol{a}$ and **OB** = $\boldsymbol{b}$.
M is the midpoint of AB and N is the point on OB such that ON : NB = 3 : 1.



Find, in terms of $\boldsymbol{a}$ and $\boldsymbol{b}$, giving each answer in its simplest form.

(a) AB

..... [1]

(b) OM

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..... [2]

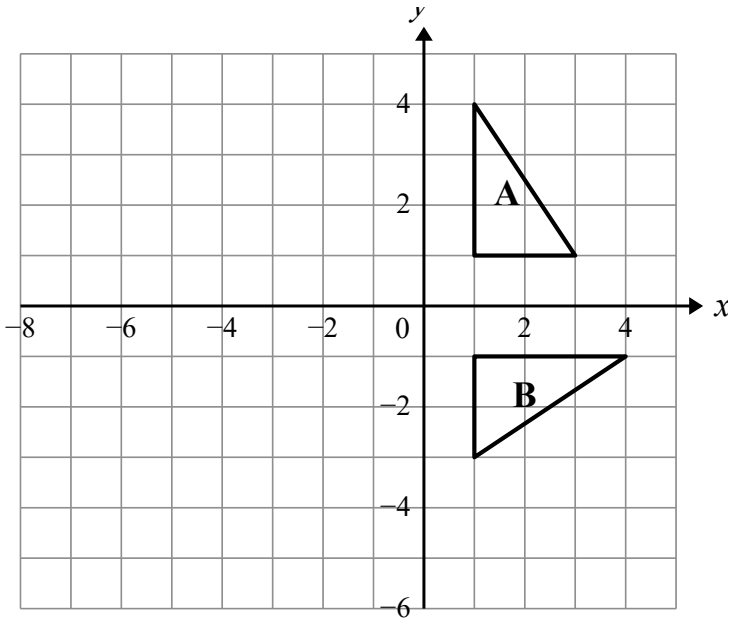
(c) MN

..... [2]

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17 The diagram shows triangle A and triangle B on a grid.



(a) Describe fully the **single** transformation that maps triangle A onto triangle B.

.....

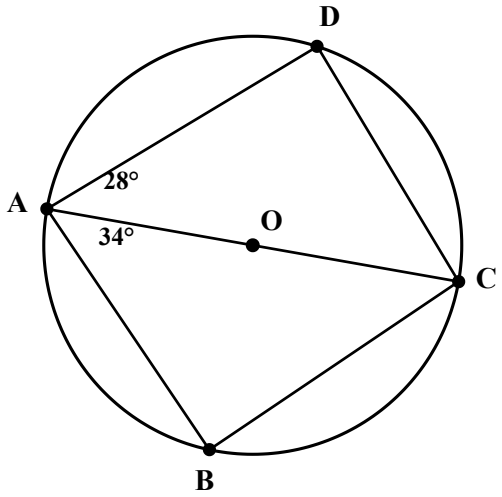
[3]

(b) On the grid, draw the image of triangle A after an enlargement with scale factor −2 and centre (0, 1).

[2]

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18 A, B, C and D lie on a circle, centre O.
AC is a diameter, angle BAC = 34° and angle CAD = 28°.



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Find the following angles, giving a geometrical reason for each answer.

(a) Angle ABC

Angle ABC = ... ° [1]

Reason ..... [1]

(b) Angle BCA

Angle BCA = ... ° [1]

(c) Angle BCD

Angle BCD = ... ° [1]

Reason ..... [1]

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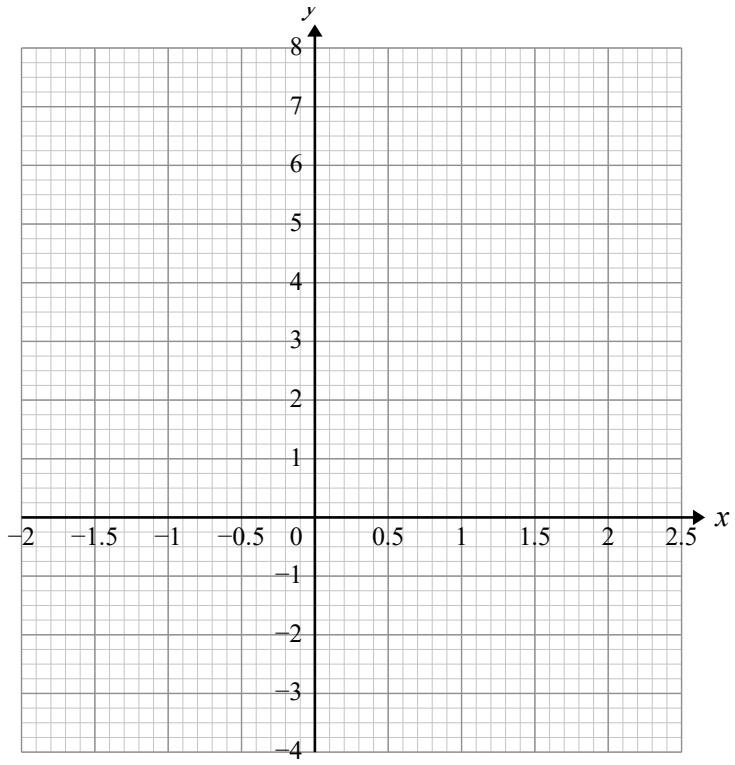
19 The table shows some values for $y = x^3 - 3x - 1$.

x	−2	−1.5	−1	−0.5	0	0.5	1	1.5	2	2.5
y	−3		1	0.375	−1	−2.375	−3		1	7.125

(a) Complete the table.

[2]

(b) On the grid, draw the graph of $y = x^3 - 3x - 1$ for $-2 \leqslant x \leqslant 2.5$.



[4]

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(c) By drawing a suitable tangent, estimate the gradient of the graph at the point where $x = 2$.

.....

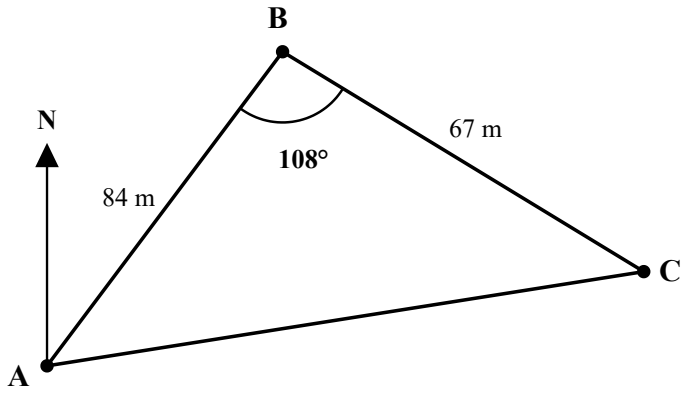
[3]

Question 20 is on page 13.

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20 A, B and C are three points on level ground.
AB = 84 m, BC = 67 m and angle ABC = 108°.



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(a) Calculate AC.

AC = ... m [4]

(b) Calculate angle BAC.

Angle BAC = ... ° [3]

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(c) The bearing of B from A is 052° and C lies to the east of the line AB.
Find the bearing of C from A.

.....° [1]

(d) A vertical mast of height 30 m stands at C.
Calculate the angle of elevation of the top of the mast from A.

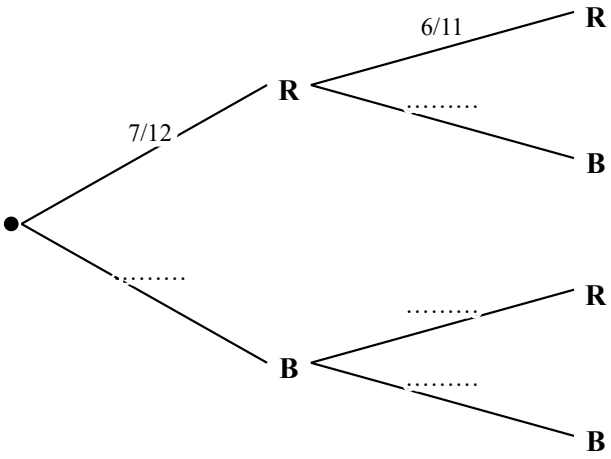
.....° [2]

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21 A bag contains 7 red counters and 5 blue counters.
Two counters are taken at random from the bag, without replacement.

(a) Complete the tree diagram.



[2]

(b) Find the probability that both counters are red.

..... [2]

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(c) Find the probability that the two counters are different colours.

.....

[2]

(d) Three counters are now taken at random from the original bag, without replacement.
Find the probability that at least one of these counters is blue.

.....

[3]

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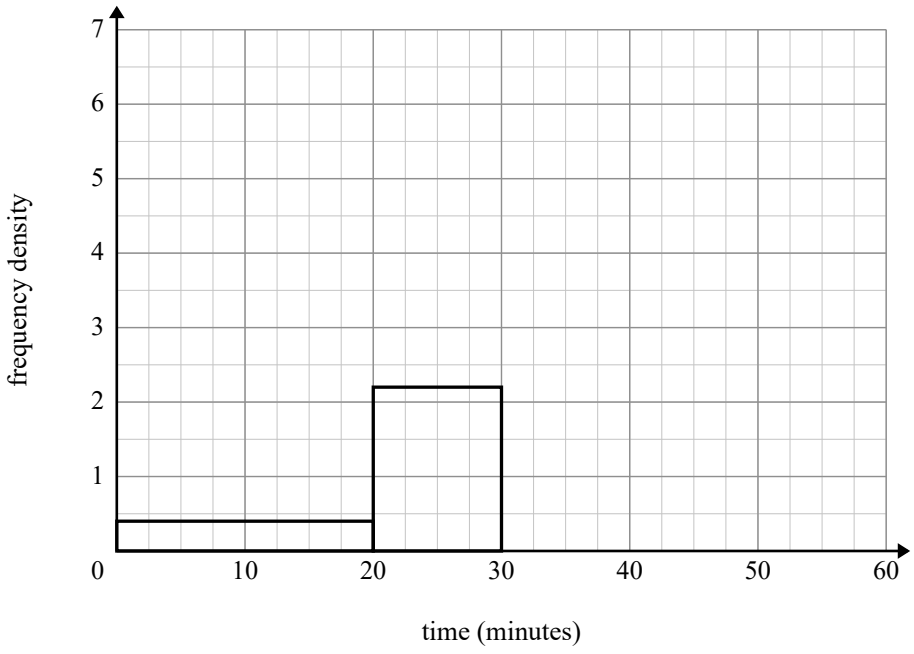
22 The table shows the times, t minutes, taken by 120 students to finish a test.

Time (t minutes)	$0 < t \leqslant 20$	$20 < t \leqslant 30$	$30 < t \leqslant 35$	$35 < t \leqslant 40$	$40 < t \leqslant 60$
Frequency	8	22	30	34	26

(a) Calculate an estimate of the mean time.

..... min [4]

(b) Complete the histogram to show this information.
The first two bars have been drawn for you.

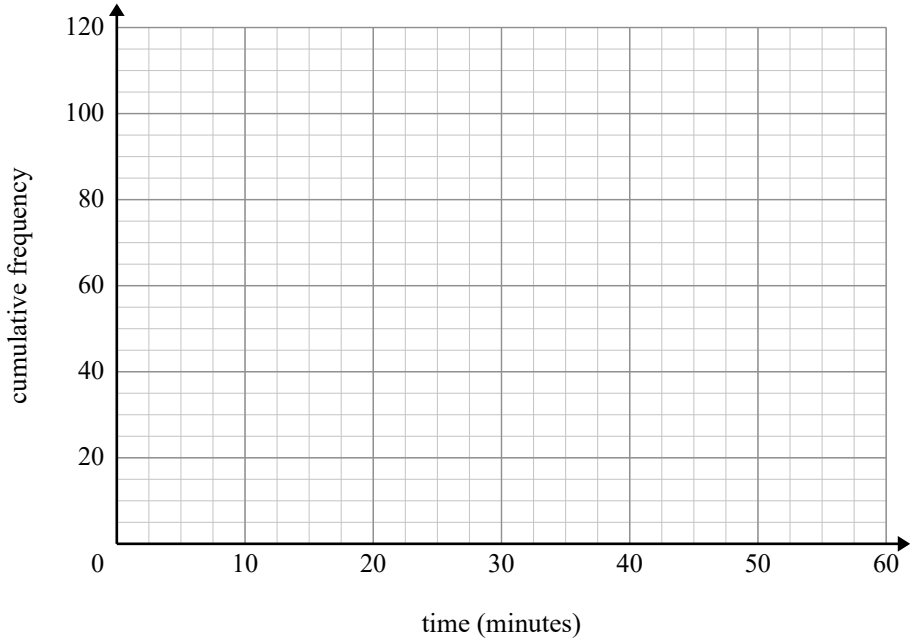


[3]

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(c) Complete the cumulative frequency table and draw the cumulative frequency diagram on the grid.

Time (t minutes)	$t \leqslant 20$	$t \leqslant 30$	$t \leqslant 35$	$t \leqslant 40$	$t \leqslant 60$
Cumulative frequency	8				120



[2]

(d) Use your diagram to estimate the interquartile range.

..... min [2]

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